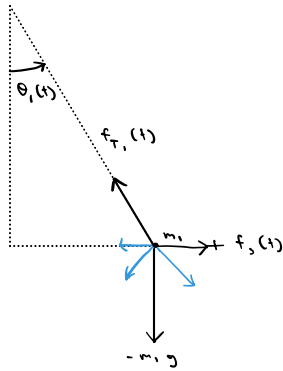


State Coupled Pendula Using Matrices

Let's analyze the forces acting on each mass in the system.

Free-body diagram for m_1

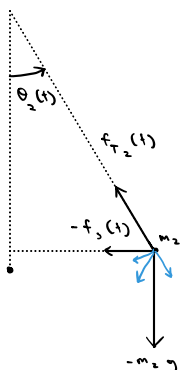


Net Forces Acting on m_1

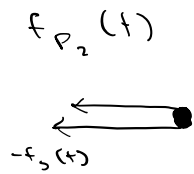
$$f_{p,1}(t) \longleftrightarrow f_s(t)$$

$$\begin{aligned} \underbrace{\sum F_i}_{\text{net forces acting on system}} &= f_{p,1}(t) + f_s(t) \\ &= \frac{-m_1 \cdot g}{l} \cdot u_1(t) + R(u_2(t) - u_1(t)) \\ &= \left(\frac{-m_1 \cdot g}{l} - k \right) u_1(t) + R \cdot u_2(t) \end{aligned}$$

Free-body diagram for m_2



Net Forces Acting on m_2



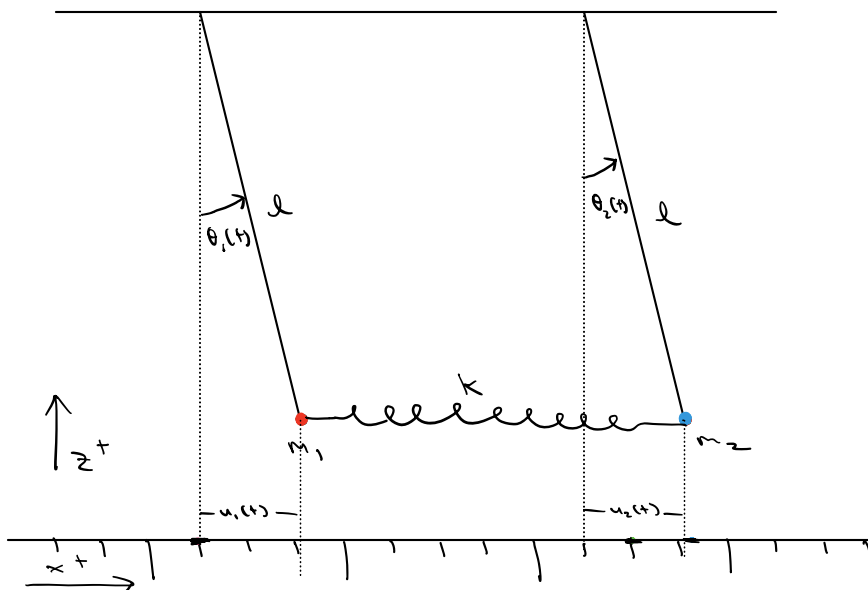
$$\begin{aligned}\sum F_2 &= f_{P_2}(t) + -f_s(t) \\ &= \frac{-m_2 \cdot g}{l} u_2(t) - R(u_2(t) - u_1(t)) \\ &= R \cdot u_1(t) + \left(-R - \frac{m_2 \cdot g}{l}\right) u_2(t)\end{aligned}$$

Now, let's derive the coupled differential equations to model the motion of the masses in the system.

Recall Newton's Second Law:

$$\sum F = m \cdot a(t)$$

This time, however, we want to use vectors about each mass in an organized format.



$$\begin{bmatrix} \sum F_1 \\ \sum F_2 \end{bmatrix} = \begin{bmatrix} m_1 \cdot \ddot{u}_1(t) \\ m_2 \cdot \ddot{u}_2(t) \end{bmatrix}$$

$$= \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} \ddot{u}_1(t) \\ \ddot{u}_2(t) \end{bmatrix}$$

$$= \underbrace{[M]}_{2 \times 2} [\ddot{u}(t)]_{2 \times 1}$$

where $\vec{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$

$$\begin{bmatrix} \sum F_1 \\ \sum F_2 \end{bmatrix} = \begin{bmatrix} \left(-\frac{m_1 \cdot g}{\ell} - R\right) u_1(t) + R u_2(t) \\ R u_1(t) + \left(-R - \frac{m_2 \cdot g}{\ell}\right) u_2(t) \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{m_1 \cdot g}{\ell} - R & R \\ R & -R - \frac{m_2 \cdot g}{\ell} \end{bmatrix}_{2 \times 2} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}_{2 \times 1}$$

$$= - \begin{bmatrix} \frac{m_1 \cdot g}{\ell} + R & -R \\ -R & R + \frac{m_2 \cdot g}{\ell} \end{bmatrix}_{2 \times 2} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}_{2 \times 1}$$

$$= \underbrace{-K}_{\text{stiffness matrix}} \cdot \vec{u}(t)$$

$$\Rightarrow M \ddot{\vec{u}}(t) = -K \vec{u}(t) \quad \leftarrow \text{matrix version of Newton's 2nd Law}$$

matrix version of simple harmonic oscillator ODE

$$\Rightarrow M \ddot{\vec{u}}(t) + K \vec{u}(t) = 0$$

$$\text{where } M = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \quad \vec{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$$

$$\text{and } K = \begin{bmatrix} \frac{m_1 \cdot g}{l} + R & -R \\ -R & R + \frac{m_2 \cdot g}{l} \end{bmatrix}$$

40 min